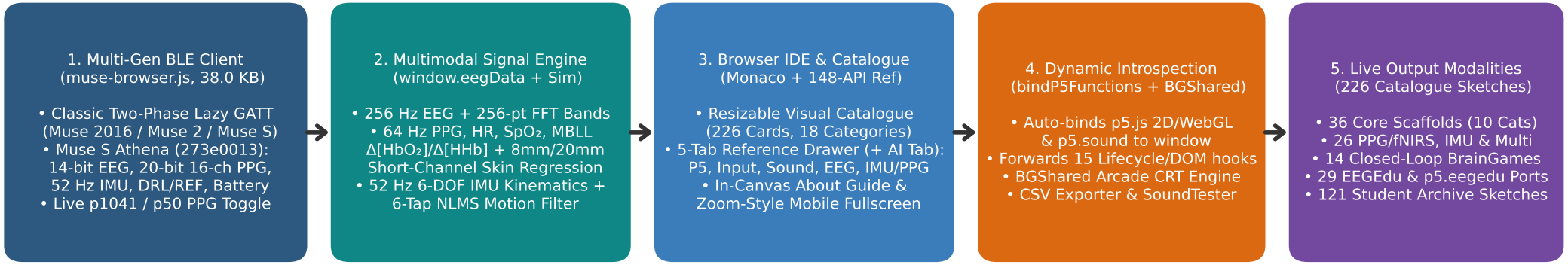
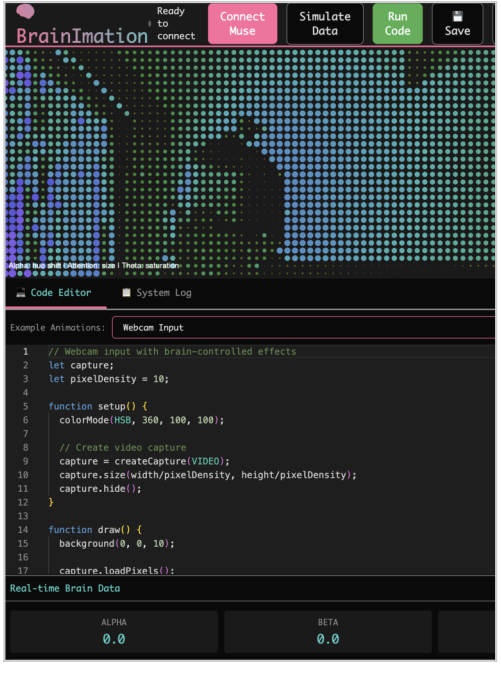
A. BrainImation Multimodal Client-Side Architecture: Muse S Athena + Classic BLE Client, PPG/fNIRS, 6-DOF IMU & Visual Catalogue



B. BrainImation Live Browser IDE & Visual Catalogue



C. Unified 18-Category Visual Catalogue (226 Interactive Sketches with Thumbnail Cards)

Catalogue Category / Track	N	Med.	Representative Sketches	Primary Physiological & API Modalities
1–3. Intro, Bands & Mindfulness	11	27	basic, bandOrbitz, mandala	eegData.alpha..gamma, attention, meditation
4–6. Particles, 3D WebGL & Draw	9	32	particles, 3dTorus, BrainDraw	WEBGL 3D torus, particle physics, brush input
7. Raw 256 Hz EEG & Online ERPs	3	44	eegtrace, multichannel, epochs	256 Hz TP9..TP10 & K=20 ERP averaging
8–10. Noise, Webcam & FM Sound	13	35	perlinNoise, webcamInput, brainSynth	Perlin fields, webcam mirror, Chowning FM FFT
11. PPG & Optical (fNIRS Studio)	10	194	fNIRSBeerLambert, ppgShortChannel...	64 Hz 660/730/850nm, HbO₂/HHb, 8/20mm OLS
12. Motion, IMU & Broadcast	11	215	imuStackedKinematics, imuHeadMove...	52 Hz Accel/Gyro, 2nd-order pos., NLMS filter
13. Multimodal (EEG + PPG + HR)	5	117	multiNeurovascularCoupling, HEP	Neurovascular lag, PPG-peak locked HEP
14. BrainGames Arcade Suite	14	514	gameGolfShooter, gameArchery, FPS	14 closed-loop BCI arcade games (BGShared)
15. EEGEdu Interactive Modules	13	131	eegeduFlock3D, Oddball, SSVEP+CSV	Per-channel PSD, oddball/SSVEP CSV export
16. Classic p5.eegedu Collection	16	37	eegeduAlphaSnake, eegeduEpochs	Preserved classic p5.eegedu creative sketches
17. 2025 Neurotech Catalogue	95	149	EegFlappyBird, EegTetris, CatErp	Curated PSYCH 403 BCI games & 4-ch ERPs
18. 2022 NeuroArt Class Gallery	26	43	studentBrainTyphoon, Terrain3D	Inaugural 2022 undergraduate neuro-art gallery